



Tecnológico de Monterrey

Project

Implementation of a Cloud-Based LLM Solution for LSM Interpretation

Previous Work

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1. Introduction

Sign languages constitute complete and natural linguistic systems used by deaf and hard-of-hearing communities worldwide. Unlike spoken languages, sign languages rely on hand gestures, facial expressions, body posture, and movement dynamics to convey meaning. American Sign Language (ASL) and Mexican Sign Language (LSM) possess their own grammatical and syntactic structures, making automatic translation and recognition highly challenging tasks [1].

Communication barriers between deaf and hearing communities continue to represent an important accessibility challenge in education, healthcare, workplaces, and public services. In recent years, advances in Artificial Intelligence (AI), Computer Vision, and Deep Learning have enabled the development of automatic sign language recognition and translation systems capable of interpreting gestures and converting them into text or speech in real time.

Research in this area has evolved from traditional sensor-based systems toward modern vision-based architectures using deep learning models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models [2]. Furthermore, recent studies have demonstrated that landmark-based approaches using pose estimation frameworks such as MediaPipe can significantly reduce computational complexity while maintaining competitive performance [3].

2. Evolution of Recognition Systems

2.1 Traditional Sign Language Recognition Systems

Early sign language recognition systems relied primarily on sensor-based technologies such as data gloves, accelerometers, inertial measurement units (IMUs), and motion capture devices. These systems provided accurate gesture tracking but suffered from several limitations, including high hardware costs, intrusive wearable components, and reduced scalability for real-world deployment [4].

Subsequently, researchers transitioned toward vision-based systems using cameras combined with conventional computer vision techniques. Traditional feature extraction methods such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and Haar Cascade classifiers were used to identify hand shapes and gestures from images and video sequences. However, these approaches were highly sensitivity to environmental conditions such as illumination, occlusion, and background complexity. The introduction of deep learning significantly improved the robustness and accuracy of sign language recognition systems [5].

2.2 Feature Extraction in the Deep Learning Era (Front-end)

In the context of real-time video processing, computational efficiency is critical. Current trends for LSM are divided into two main approaches:

- **Pixel-Based Approaches (RGB):** These utilize CNNs such as EfficientNet or MobileNetV3 to extract visual features directly from video frames. While they

capture fine textures and environmental context, they are often computationally expensive for mobile deployment and sensitive to lighting changes.

- **Skeleton-Based Approaches (Keypoints):** This is the current gold standard for real-time applications. Tools like MediaPipe allow for the extraction of 2D/3D coordinates of hand joints, face landmarks, and torso posture [3]. This results in a drastic reduction in dimensionality—from millions of pixels to a few hundred coordinates—and enhances user privacy by avoiding the storage of raw video.

3. Temporal Dependency Modeling (Back-end)

LSM is not a series of static images but a temporal sequence where meaning depends on both previous and subsequent contexts. The modeling of these sequences has shifted significantly:

- **Recurrent Architectures (RNN/LSTM/GRU):** Historically used to handle sequences, these models suffer from the "vanishing gradient" problem in long sentences and are difficult to parallelize, limiting their efficiency during the training of large LSM datasets [2].
- **Transformers and Attention Mechanisms:** The state of the art has migrated toward Transformers. Through the Self-Attention mechanism, models can identify which specific video frames are crucial for understanding a particular sign, regardless of their position in the sequence [1].
- **Specialized Graphs:** Since body points naturally form a graph structure, Spatial-Temporal Graph Convolutional Networks (ST-GCN) are increasingly used to model the biological relationship between joints (e.g., the connection between the wrist and fingers) before the translation stage [6].

4. The Data Challenge in LSM

Unlike ASL, LSM lacks large-scale, publicly labeled datasets. Researchers are currently addressing this scarcity through:

- **Data Augmentation:** Applying rotations, scaling, and "keypoint jittering" to synthetic or limited samples.
- **Transfer Learning:** Pre-training models on ASL or Spanish Sign Language (LSE) and performing fine-tuning on the LSM Corpus or proprietary databases from institutions like IPN or UNAM [7]. Recent efforts in Mexico have focused on creating robust datasets like the LSM-25 to bridge this gap [8].

5. Real-Time Processing and Implementation

To achieve the latency required for fluid conversation (typically $< 100\text{ms}$), current research focuses on:

- **Model Quantization:** Reducing AI weight precision from `float32` to `int8` to enable inference on edge devices.

- Knowledge Distillation: Using a heavy "teacher" Transformer to train a lightweight "student" model optimized for web browsers via TensorFlow.js or ONNX Runtime [9].

6. Large Language Models in Sign Language Translation

Recent advances in Large Language Models (LLMs) have opened new possibilities for sign language interpretation systems. Traditional sign language recognition models primarily focus on gesture classification; however, they often struggle to generate grammatically coherent natural language sentences. This limitation is especially important because sign languages possess grammatical structures that differ significantly from spoken languages such as English or Spanish [1].

LLMs provide a mechanism for contextual understanding and sentence generation. After a gesture recognition model predicts glosses or intermediate symbolic representations, an LLM can be used to transform those outputs into fluent natural language text. This process resembles neural machine translation, where semantic meaning is preserved while adapting the grammatical structure to the target language [11].

Transformer-based architectures such as GPT, BERT, T5, and BART have demonstrated strong capabilities in sequence-to-sequence tasks, contextual inference, and language generation [1][11]. Researchers are increasingly integrating these architectures into sign language pipelines to improve semantic coherence and contextual interpretation.

Another important advantage of LLMs is their ability to manage ambiguity. Many signs may possess multiple meanings depending on context, facial expressions, or sentence position. By leveraging contextual embeddings and attention mechanisms, LLMs can infer more accurate sentence interpretations compared to isolated classification systems [12].

Furthermore, cloud-based LLM deployment allows scalable inference and centralized model updates. This enables educational institutions and accessibility platforms to continuously improve translation quality without requiring users to update local hardware. Nevertheless, integrating LLMs introduces challenges such as computational cost, latency, and the need for large multilingual datasets [9].

The integration of LLMs into sign language interpretation systems represents a major transition from gesture recognition toward semantic communication systems capable of contextual understanding and natural interaction.

7. Cloud Computing and Scalable AI Architectures

Cloud computing has become a fundamental component in modern Artificial Intelligence systems, particularly in applications requiring high computational performance, scalability, and real-time processing. In the context of sign language interpretation, cloud-based infrastructures allow deep learning models to process video streams efficiently while reducing the computational burden on local devices [13].

Platforms such as Google Cloud Platform (GCP), Microsoft Azure, and Amazon Web Services (AWS) provide specialized AI services capable of supporting computer vision inference, GPU acceleration, distributed training, and large-scale storage systems. These services are especially useful for educational accessibility projects because they facilitate collaborative development and scalable deployment.

Cloud architectures also simplify the integration of multiple AI modules. For example, a complete sign language interpretation pipeline may include:

- Video capture from a webcam or mobile device.
- Landmark extraction using MediaPipe [3].
- Temporal sequence modeling using Transformers or LSTMs [1][2].
- Natural language post-processing using LLMs [11].
- Text-to-speech generation for auditory output.

Using cloud computing, each component can operate as an independent microservice connected through APIs. This modular architecture improves maintainability, scalability, and future extensibility.

Another important advantage of cloud deployment is centralized dataset management. Since LSM datasets remain limited, cloud storage enables collaborative annotation and continuous dataset expansion across institutions and research groups [8]. Additionally, cloud-based pipelines facilitate distributed model training using GPUs or TPUs, reducing training time for computationally intensive architectures.

Despite these benefits, cloud-based systems also introduce concerns related to privacy, bandwidth usage, latency, and operational costs. Researchers must carefully evaluate trade-offs between local edge inference and cloud-based processing to achieve optimal performance in educational environments.

8. Challenges in Mexican Sign Language Recognition

Although significant progress has been achieved in sign language recognition, Mexican Sign Language (LSM) continues to face several unique challenges that limit the development of robust AI systems.

One of the primary challenges is the scarcity of publicly available datasets. Compared to ASL, which benefits from large annotated corpora, LSM datasets remain relatively small and fragmented [8]. This limitation affects the ability of deep learning models to generalize across different users, environments, and communication styles.

Another challenge involves regional variation. LSM signs may differ depending on geographic location, educational institution, or social community. As a result, a gesture recognized correctly in one region may possess a different interpretation elsewhere. This linguistic variability complicates model standardization and dataset labeling.

Co-articulation also represents a significant technical problem. Similar to spoken language, signs are influenced by neighboring gestures and motion continuity. Consequently, isolated sign classification is insufficient for understanding continuous sentences. Temporal modeling and contextual interpretation become essential components for accurate recognition [1][2].

Environmental variability introduces additional complications. Lighting conditions, occlusions, camera quality, and background complexity can negatively affect landmark extraction and gesture recognition performance. Although MediaPipe significantly improves robustness, real-world deployment remains challenging in uncontrolled environments [3].

Another important issue is the grammatical gap between LSM and spoken Spanish. Sign languages are not direct visual representations of spoken language; instead, they possess their own syntax, spatial grammar, and semantic structures. Therefore, translation systems must perform semantic adaptation rather than literal word substitution [1].

Finally, ethical considerations must also be addressed. AI systems designed for accessibility should prioritize inclusivity, privacy, transparency, and bias mitigation. Since many landmark-based approaches avoid storing raw video, they provide an additional advantage in terms of user privacy and responsible AI deployment [13].

9. Educational and Social Impact

The implementation of intelligent sign language interpretation systems has the potential to generate substantial educational and social benefits. Communication barriers continue to affect academic inclusion, employment opportunities, healthcare accessibility, and participation in public services for deaf individuals worldwide [10].

Educational institutions can particularly benefit from AI-assisted interpretation systems. Real-time translation tools may support communication between students and instructors during lectures, online learning environments, and collaborative

activities. Additionally, interactive educational platforms may help hearing individuals learn basic sign language skills, promoting greater inclusion and social awareness.

In healthcare environments, automatic interpretation systems may facilitate communication between patients and medical personnel, reducing misunderstandings and improving access to medical information. Similar applications can be extended to customer service, government institutions, and workplace environments.

From a technological perspective, lightweight AI architectures optimized for mobile deployment increase accessibility by enabling users to access interpretation tools through smartphones and tablets [9]. This democratization of AI technologies is especially important in regions where specialized accessibility infrastructure remains limited.

Furthermore, cloud-based systems can continuously improve through centralized updates, collaborative dataset expansion, and model retraining. As additional LSM datasets become available, recognition accuracy and contextual understanding are expected to improve significantly over time [8].

The convergence of Artificial Intelligence, Computer Vision, and Natural Language Processing therefore represents a promising opportunity for reducing communication barriers and fostering greater digital inclusion for deaf communities [1][10].

10. Conclusions

The field of sign language recognition and translation has experienced significant evolution in recent years, transitioning from traditional sensor-based systems toward modern Artificial Intelligence architectures based on Computer Vision, Deep Learning, and Natural Language Processing. Current research trends demonstrate a clear movement toward End-to-End systems that prioritize lightweight, landmark-based approaches capable of operating in real time while maintaining competitive accuracy [1][3].

The integration of technologies such as MediaPipe, Transformer-based architectures, Spatial-Temporal Graph Neural Networks, and Large Language Models has substantially improved the ability of AI systems to interpret complex temporal gesture sequences and generate more coherent natural language outputs [1][6][11]. These advances are especially relevant for educational and accessibility applications, where real-time interaction and scalability are critical requirements.

Despite this progress, Mexican Sign Language (LSM) continues to face important challenges, including limited publicly available datasets, regional linguistic variations, co-articulation effects, and the grammatical differences between sign language and spoken Spanish [8]. These limitations highlight the necessity for additional research focused on dataset creation, multimodal temporal modeling, and contextual language understanding specifically adapted to LSM environments.

Furthermore, the emergence of cloud computing and lightweight AI optimization techniques creates new opportunities for deploying scalable and accessible interpretation systems on mobile devices and web-based platforms [9][13]. This democratization of AI technologies may contribute significantly toward reducing communication barriers in educational institutions, healthcare services, workplaces, and public environments.

Overall, the reviewed literature demonstrates that the convergence of Artificial Intelligence, Computer Vision, cloud infrastructures, and Large Language Models provides a strong technological foundation for the development of next-generation sign language interpretation systems. In this context, the proposed project represents a feasible and relevant contribution toward improving accessibility and promoting digital inclusion for the deaf community in Mexico through intelligent real-time LSM interpretation systems.

11. References

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